

MUTUAL FUND ANALYTICS

Capstone Project I

Final Report

Integrated ETL, EDA, performance analytics, advanced risk analysis and interactive Power BI reporting

Deliverable	D7: Final Report
Dashboard	Power BI, 4 interactive pages
Advanced analytics	VaR/CVaR, rolling Sharpe, cohorts, SIP continuity, recommender, HHI
Fund universe	40 schemes

Prepared from the project notebooks, processed datasets, reports and completed Power BI dashboard.

1. Executive Summary

The Mutual Fund Analytics project combines structured mutual fund, NAV, AUM, SIP, investor transaction, portfolio holding and benchmark data into an analytical workflow. The project covers data preparation, exploratory analysis, performance measurement, risk analysis, investor behavior and an interactive Power BI dashboard.

The analytical universe contains 40 schemes. The completed dashboard reports an average 1-year return of 14.38% and an average expense ratio of 1.24%. The performance analysis identifies ICICI Pru Liquid Fund - Regular - Growth as the highest-Sharpe fund in the available scorecard, with a Sharpe ratio of 7.68.

Historical 95% downside analysis identifies SBI Small Cap Fund - Direct Plan - Growth as having the most negative VaR in the report. Its VaR is -2.69% and CVaR is -3.24% per day.

The advanced analytics layer also evaluates investor cohorts, SIP continuity, a risk-appetite recommender and sector concentration. The recommender matches Low, Moderate or High risk appetite to the corresponding risk grade and ranks matching funds by Sharpe ratio.

Area	Key result
Fund universe	40 schemes
Average 1Y return	14.38%
Average expense ratio	1.24%
Highest Sharpe	ICICI Pru Liquid Fund - Regular - Growth (7.68)
Highest downside risk	SBI Small Cap Fund - Direct Plan - Growth (VaR -2.69%)
Largest SIP cohort	2024 (■225.81 crore)
SIP at-risk rate	99.9% of eligible investors

2. Project Objectives

The project was designed to answer five practical questions:

- How large and diversified is the mutual fund universe?
- Which funds and categories show stronger return and risk-adjusted performance?
- What downside risk is visible in historical NAV returns?
- How do investor transactions and SIP behavior vary across cohorts?
- How can analytics be exposed through an interactive dashboard and a simple risk-based recommender?

3. Data Coverage

The processed project contains 10 main analytical datasets covering fund master data, NAV history, AUM by fund house, monthly SIP inflows, category inflows, industry folio counts, scheme performance, investor transactions, portfolio holdings and benchmark indices.

Dataset	Rows	Primary purpose
01 Fund master	40	Scheme metadata and classifications
02 NAV history	64,320	Historical NAV and return calculations
03 AUM by fund house	90	AUM and scheme-count trends
04 Monthly SIP inflows	48	SIP flow and account trends
05 Category inflows	144	Category-level net inflows
06 Industry folio count	21	Folio growth by asset class
07 Scheme performance	40	Return, alpha, beta, Sharpe, Sortino and risk
08 Investor transactions	32,778	Investor and transaction behavior
09 Portfolio holdings	322	Sector concentration analysis
10 Benchmark indices	8,050	Market benchmark comparison

4. Methodology

ETL and data preparation

The project follows a structured ingestion and processing flow. Raw datasets are organized separately from processed datasets, and the processed files are used by the analytical notebooks and dashboard. The project structure also separates scripts, SQL, notebooks, dashboard assets and reports.

Performance analytics

The performance dataset contains 1-year, 3-year and 5-year returns together with benchmark return, alpha, beta, Sharpe ratio, Sortino ratio, annualized standard deviation, maximum drawdown, AUM, expense ratio, rating and risk grade.

Risk analytics

Daily NAV returns are calculated as $\text{NAV}(t) / \text{NAV}(t-1) - 1$. Historical VaR at 95% confidence is the 5th percentile of daily returns, while CVaR is the mean of observations at or below the VaR threshold.

The rolling Sharpe analysis uses a 90-day window and annualizes using $\sqrt{252}$. The project therefore uses trading-day annualization rather than calendar-day annualization.

Investor analytics

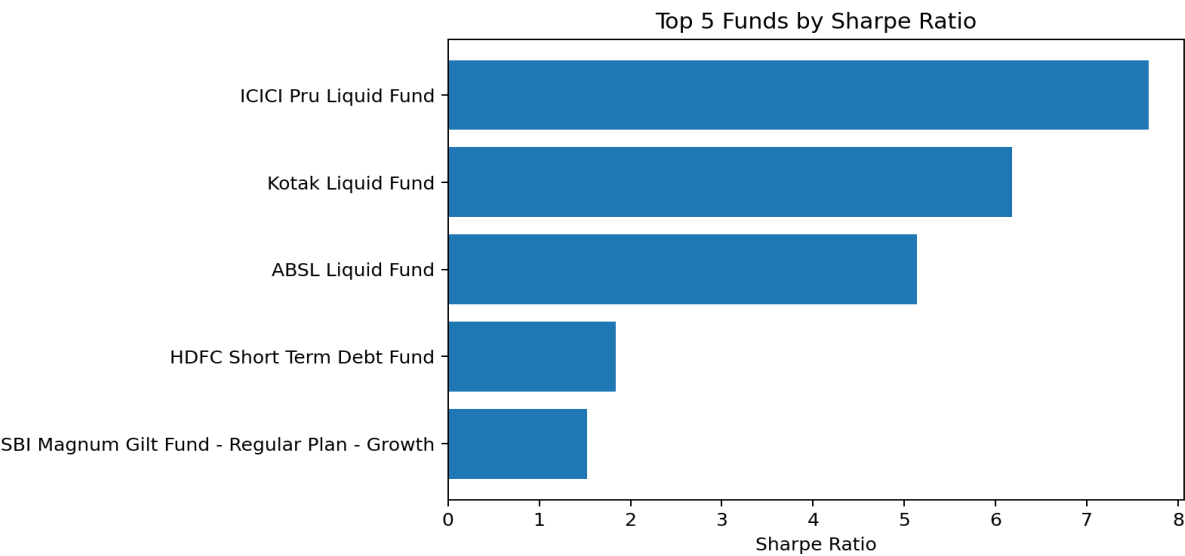
Investor cohort analysis groups investors by cohort year and measures SIP activity. SIP continuity analysis includes investors with at least six SIP transactions and flags an investor as at-risk when the maximum observed gap between SIP dates exceeds 35 days.

Concentration and recommendation

Sector HHI is calculated as the sum of squared sector weights after converting percentages to decimals. The recommender accepts Low, Moderate or High risk appetite, filters funds by matching risk grade and returns the top three by Sharpe ratio.

5. Exploratory and Performance Findings

The completed Power BI and analytical outputs show meaningful differences between fund categories and risk grades. Small-cap and mid-cap schemes appear prominently among the higher-return funds, while liquid and short-duration funds dominate the highest Sharpe values because their return volatility is lower in the available scorecard.

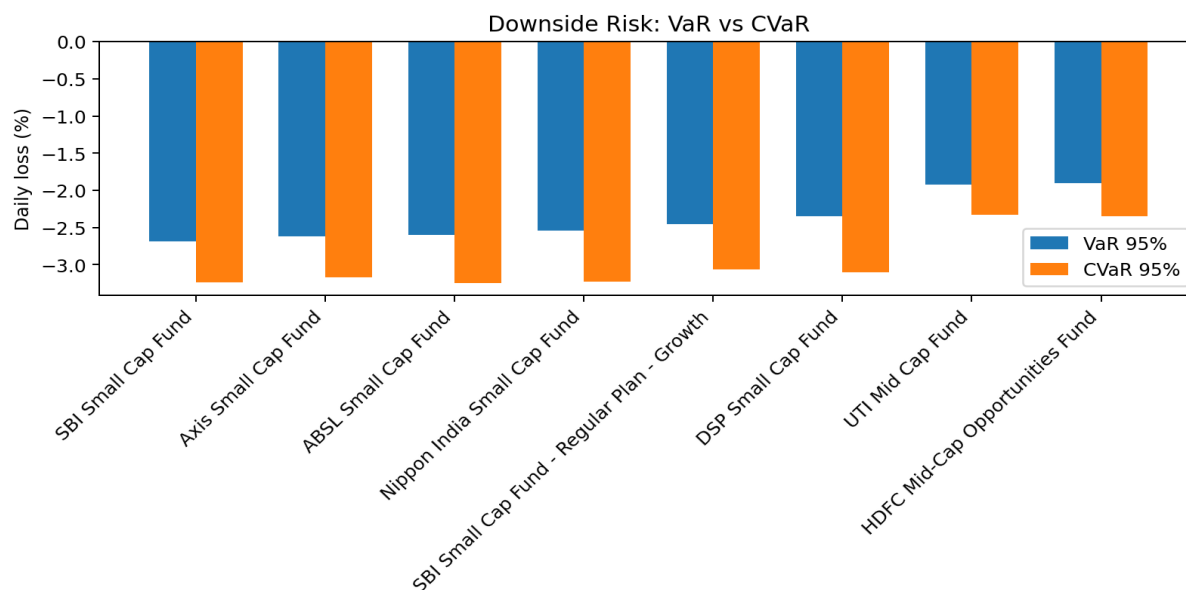


Among the five highest Sharpe scores, ICICI Pru Liquid Fund - Regular - Growth leads at 7.68, followed by Kotak Liquid Fund - Regular - Growth at 6.18.

The dashboard's Fund Performance page supports interactive comparison using Fund House, Category and Plan filters, along with Risk Grade analysis, return comparisons and a detailed fund table.

6. Risk Analysis

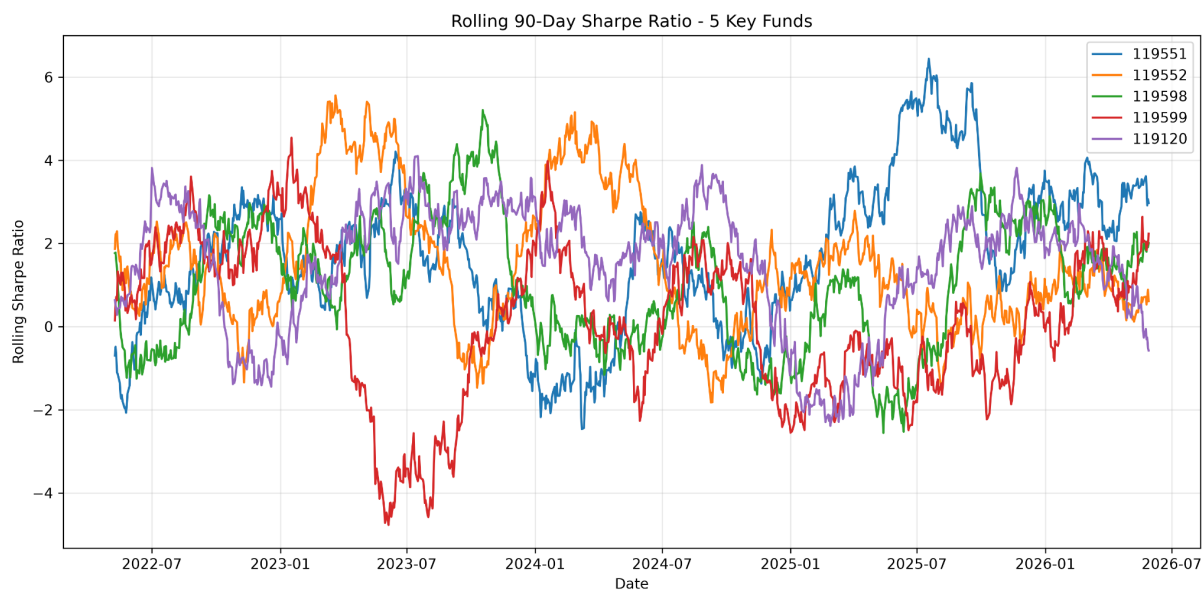
The historical VaR/CVaR analysis covers the 40-scheme universe. The most negative VaR belongs to SBI Small Cap Fund - Direct Plan - Growth, with a 95% daily VaR of -2.69% and CVaR of -3.24%. Small-cap funds form the most visible group among the highest downside-risk observations.



Interpretation: VaR identifies a percentile threshold for adverse daily returns, while CVaR gives the average loss in the tail beyond that threshold. CVaR is therefore useful when comparing the severity of extreme downside outcomes.

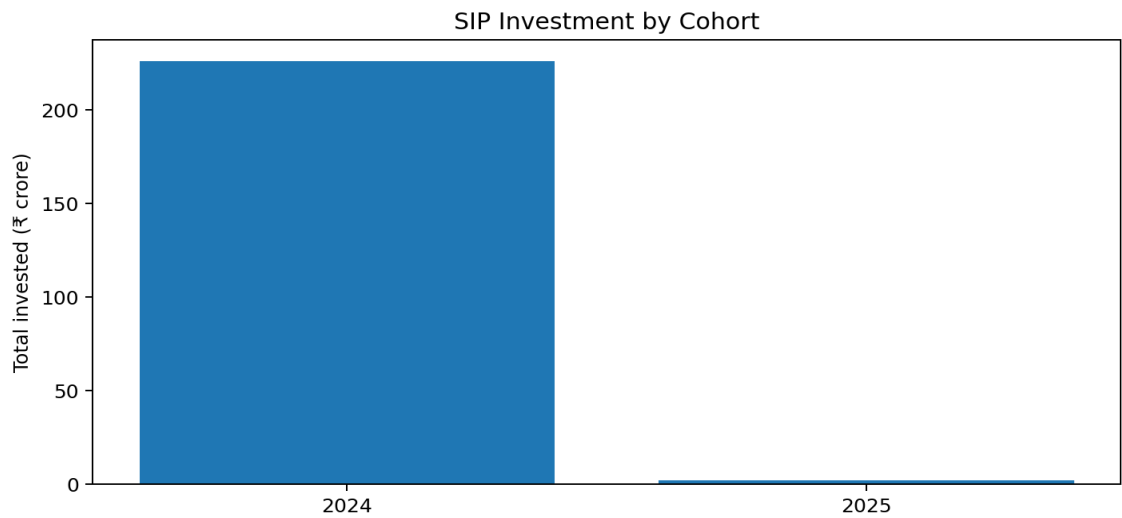
7. Advanced Analytics

Rolling 90-Day Sharpe



The rolling chart tracks changing risk-adjusted performance for five key funds. It shows that risk-adjusted performance is time-varying rather than constant, with several funds moving between positive and negative rolling Sharpe periods.

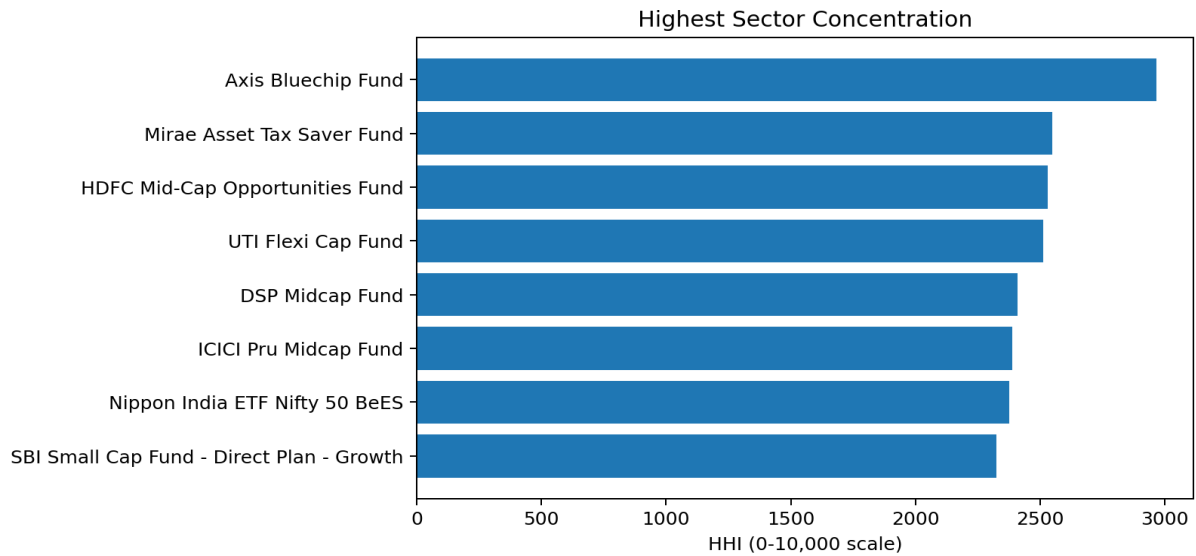
Investor cohort and SIP continuity



The largest SIP investment cohort in the available report is 2024, with total invested amount of approximately █225.81 crore. The cohort's average SIP amount is approximately █10,997.

8. Concentration and Recommendation Logic

The highest sector HHI in the available equity holdings is for Axis Bluechip Fund - Regular - Growth, with HHI 0.2968, equivalent to 2968 on the 0-10,000 scale. The project classifies this observation as high concentration.



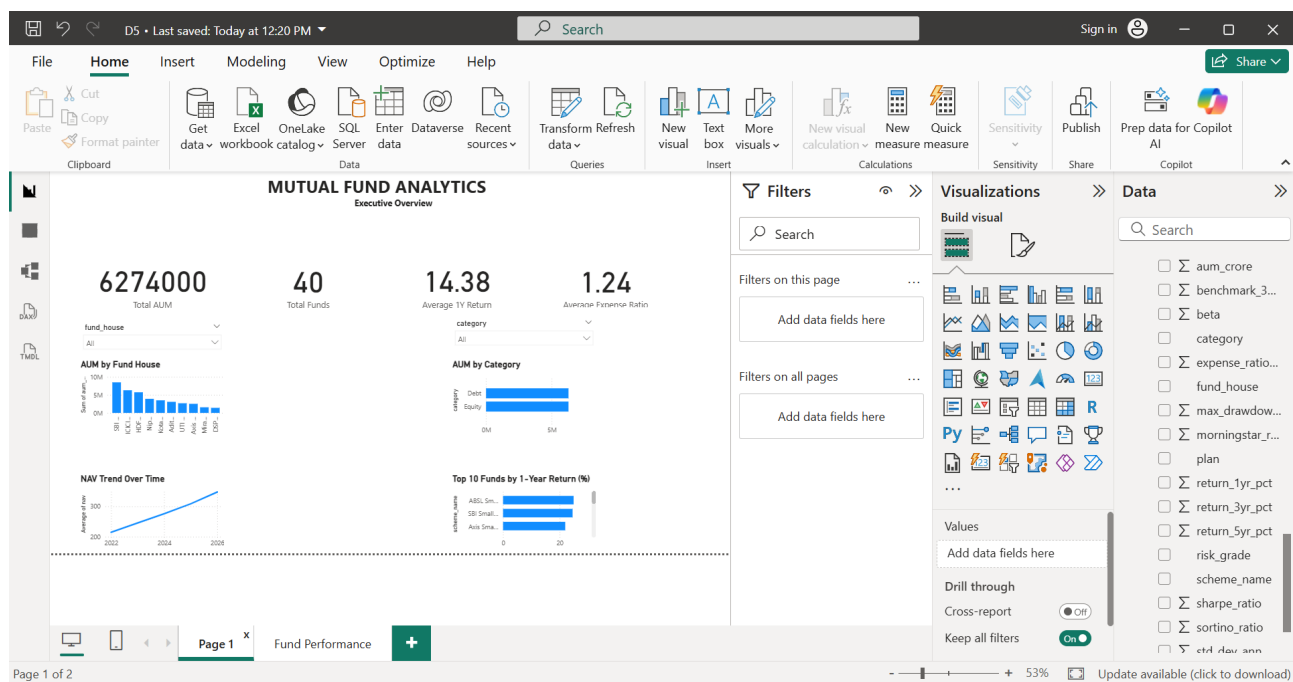
Risk-appetite recommender

The recommender supports three inputs: Low, Moderate and High. It normalizes the requested risk appetite, filters funds whose risk grade matches the requested level, sorts by Sharpe ratio and returns the top three funds. This creates a simple explainable recommendation rule rather than a black-box model.

SIP continuity analysis identifies 99.9% of eligible investors as at-risk under the project's >35-day maximum-gap rule. This result should be interpreted as a project-specific behavioral flag rather than a prediction of investor churn.

9. Interactive Power BI Dashboard

The final dashboard contains four pages: Executive Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The pages use KPI cards, charts, tables and interactive slicers to support exploration from portfolio-level views to fund-level and investor-level details.



Page	Purpose
Executive Overview	AUM, fund count, average return, expense ratio and high-level trends
Fund Performance	Return, risk-adjusted metrics, fund comparisons and risk filters
Investor Analytics	Investor counts, transaction activity, demographics and payment modes
SIP & Market Trends	SIP inflows, category trends and benchmark performance

10. Key Findings and Recommendations

Key findings

1. ICICI Pru Liquid Fund - Regular - Growth has the highest Sharpe ratio in the available performance scorecard at 7.68.
2. SBI Small Cap Fund - Direct Plan - Growth has the highest historical downside risk in the VaR/CVaR report, with VaR -2.69% and CVaR -3.24%.
3. The 2024 cohort has the largest total SIP investment in the available cohort report, approximately ₹225.81 crore.
4. The SIP continuity rule flags 99.9% of eligible investors as at-risk because their maximum observed SIP gap exceeds 35 days.
5. Axis Bluechip Fund - Regular - Growth has the highest sector concentration HHI among the available equity holdings at 2968.

Recommendations

- Use risk-adjusted measures such as Sharpe and Sortino alongside raw returns when comparing funds.
- Monitor high-VaR small-cap and mid-cap schemes separately because their downside tails are larger in the historical sample.
- Use SIP continuity flags as a retention-monitoring signal and investigate long gaps before drawing conclusions about churn.
- Consider sector concentration when evaluating equity-fund diversification, especially for funds with high HHI.
- Use the recommender as an explainable screening tool and combine it with investor objectives, horizon and suitability checks before any real investment decision.
- Refresh NAV and market-linked data regularly so the dashboard and risk metrics remain current.

11. Limitations

The analysis is historical and depends on the supplied project datasets. VaR and CVaR are backward-looking and do not guarantee future losses. The recommender is a rule-based screening mechanism, not personalized financial advice. The SIP at-risk percentage is highly sensitive to the project's 35-day gap definition and the six-transaction eligibility rule.

12. Conclusion

The project delivers an end-to-end mutual fund analytics workflow that connects data preparation with investment performance, risk measurement, investor behavior and interactive reporting. The strongest part of the solution is the combination of descriptive dashboarding with explainable advanced analytics such as historical VaR/CVaR, rolling Sharpe, cohort analysis, SIP continuity, HHI and risk-based recommendations.

Appendix A. Deliverable Mapping

ID	Deliverable	Status
D1	ETL pipeline script	Completed
D2	SQLite database	Completed
D3	EDA notebook	Completed
D4	Performance metrics notebook + CSVs	Completed
D5	Interactive Power BI dashboard	Completed, 4 pages
D6	Advanced analytics notebook + VaR/CVaR + recommender + rolling Sharpe	Completed
D7	Final report + presentation	This deliverable

Appendix B. Source Artifacts

Primary source artifacts used for this report: processed project datasets, 03_eda_analysis.ipynb, 04_performance_analytics.ipynb, 05_advanced_analytics.ipynb, Power BI dashboard outputs, var_cvar_report.csv, recommender.py and rolling_sharpe_chart.png.